

- Recall that a function f from the graph $G_1 = (V_1, E_1)$ to $G_2 = (V_2, E_2)$ is an *isomorphism* if f is a bijection and for all $x, y \in V_1$, we have $\{x, y\} \in E_1 \Leftrightarrow \{f(x), f(y)\} \in E_2$.

Show that if $f : G_1 \rightarrow G_2$ is an isomorphism, then there is an isomorphism $g : G_2 \rightarrow G_1$.

- Recall that two graphs G_1, G_2 are *isomorphic* if there is an isomorphism between them.

Also recall from the last worksheet the set $\mathcal{C}$, the set of all binary sequences, and how if a set $S \subseteq \mathcal{C}$ contains $()$, then S is closed downward iff $G = (S, E_{\mathcal{C}})$ is a tree, where $\{s, t\} \in E_{\mathcal{C}}$ iff s extends t by one bit or t extends s by one bit. In this problem we will call a tree like this a *Cantor tree*.

Show that if G is a tree and every vertex in G has degree at most 3, then G is isomorphic to a Cantor tree. (hint: first construct a function from the vertex set of G into $\mathcal{C}$, and then set S as the range of your function)

- Let $\mathcal{N}$ be the set of all finite sequences of positive integers. For $s, t \in \mathcal{N}$, define $\{s, t\} \in E_{\mathcal{N}}$ iff either s adds a positive integer to the end of t , or t adds a positive integer to the end of s . Just like you proved in the last worksheet for the set $\mathcal{C}$, if a subset $S \subseteq \mathcal{N}$ contains the empty sequence $()$, then S is a tree iff S is closed downward. We will call trees like this *Baire trees*.

Prove that every tree is isomorphic to a Baire tree.

- For each of the following graphs, either draw an Eulerian circuit or show that no Eulerian circuit exists.

